# Homework 1

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# 1 Question 1

## 1.1 Answer

Using the $k$-NN algorithm with $k = 3$ to find the 3 closest training instances for each test height and calculate the average of their weights.

| Person | Height ($x$) | Weight ($y$) |
|---|---|---|
| 1 | 171 cm | 80 kg |
| 2 | 168 cm | 78 kg |
| 3 | 191 cm | 100 kg |
| 4 | 182 cm | 80 kg |
| 5 | 150 cm | 65 kg |
| 6 | 178 cm | 83 kg |

**a)Test Height = 150 cm**

Absolute differences in height $|x - 150|$:

Person 1: $|171 - 150| = 21$ Person 2: $|168 - 150| = 18$ Person 3: $|191 - 150| = 41$ Person 4: $|182 - 150| = 32$ Person 5: $|150 - 150| = 0$ Person 6: $|178 - 150| = 28$

The 3 nearest neighbors are Persons 1, 2, and 5.

$$\hat{y}_{KNN} = \frac{65 + 78 + 80}{3} = \frac{223}{3} \approx 74.33kg$$

**b)Test Height = 155 cm**

Absolute differences in height $|x - 155|$:

Person 1: $|171 - 155| = 16$ Person 2: $|168 - 155| = 13$ Person 3: $|191 - 155| = 36$ Person 4: $|182 - 155| = 27$ Person 5: $|150 - 155| = 5$ Person 6: $|178 - 155| = 23$

The 3 nearest neighbors are Persons 1, 2, and 5.

$$\hat{y}_{KNN} = \frac{65 + 78 + 80}{3} = \frac{223}{3} \approx 74.33kg$$

**c)Test Height = 165 cm**

Absolute differences in height $|x - 165|$:

Person 1: $|171-165| = 6$ Person 2: $|168-165| = 3$ Person 3: $|191-165| = 26$ Person 4: $|182 - 165| = 17$ Person 5: $|150 - 165| = 15$ Person 6: $|178 - 165| = 13$

The 3 nearest neighbors are Persons 1, 2, and 6.

$$\hat{y}_{KNN} = \frac{83 + 78 + 80}{3} = \frac{241}{3} \approx 80.33kg$$

**d)Test Height = 190 cm**

Absolute differences in height $|x - 190|$:

Person 1: $|171 - 190| = 19$ Person 2: $|168 - 190| = 22$ Person 3: $|191 - 190| = 1$ Person 4: $|182 - 190| = 8$ Person 5: $|150 - 190| = 40$ Person 6: $|178 - 190| = 12$

The 3 nearest neighbors are Persons 3, 4, and 6.

$$\hat{y}_{KNN} = \frac{83 + 100 + 80}{3} = \frac{263}{3} \approx 87.67kg$$

# 2 Question 2

## 2.1 Answer

**Test Height = 150 cm**
The nearest neighbor is Person 5 ($x_5 = 150$ cm) with a distance $d = 0$. By definition, the weight becomes singular, and the prediction directly equals the label of the exact match.
$\hat{y}_{KNN} = 65.00kg$

**Test Height = 155 cm**
The 3 nearest neighbors are Persons 1, 2, and 5:

Person 1: $d_3 = |171 - 155| = 16 w_3 = \frac{1}{16} = 0.0625$ Person 2: $d_2 = |168 - 155| = 13 w_2 = \frac{1}{13} \approx 0.0769$ Person 5: $d_1 = |150 - 155| = 5 w_1 = \frac{1}{5} = 0.2$

$\hat{y}_{KNN} = \frac{(0.2\times65)+(0.0769\times78)+(0.0625\times80)}{0.2+0.0769+0.0625} = \frac{23.9982}{0.3394} \approx 70.71kg$

**Test Height = 165 cm**

The 3 nearest neighbors are Persons 1, 2, and 6:

Person 1: $d_2 = |171 - 165| = 6, \quad w_2 = \frac{1}{6} \approx 0.1667$ Person 2: $d_1 = |168 - 165| = 3, \quad w_1 = \frac{1}{3} \approx 0.3333$ Person 6: $d_3 = |178 - 165| = 13, \quad w_3 = \frac{1}{13} \approx 0.0769$

$\hat{y}_{KNN} = \frac{(0.3333\times78)+(0.1667\times80)+(0.0769\times83)}{0.3333+0.1667+0.0769} = \frac{45.7187}{0.5769} \approx 79.25kg$

**Test Height = 190 cm**

The 3 nearest neighbors are Persons 3, 4, and 6:

Person 3: $d_1 = |191 - 190| = 1, \quad w_1 = \frac{1}{1} = 1.0$ Person 4: $d_2 = |182 - 190| = 8, \quad w_2 = \frac{1}{8} = 0.125$ Person 6: $d_3 = |178 - 190| = 12, \quad w_3 = \frac{1}{12} \approx 0.0833$

$\hat{y}_{KNN} = \frac{(1.0\times100)+(0.125\times80)+(0.0833\times83)}{1.0+0.125+0.0833} = \frac{116.9139}{1.2083} \approx 96.76kg$

# 3 Question 3

## 3.1 Answer

Given $J(\mathbf{x}) = \mathbf{x}^T\mathbf{Q}\mathbf{x} + \mathbf{d}^T\mathbf{x} + c$, where $\mathbf{Q} = \mathbf{Q}^T$.

### a) Gradient $\nabla_{\mathbf{x}} J(\mathbf{x}) = 2\mathbf{Q}\mathbf{x} + \mathbf{d}$

The linearity of differentiation:

$$\nabla_{\mathbf{x}} J(\mathbf{x}) = \nabla_{\mathbf{x}}(\mathbf{x}^T\mathbf{Q}\mathbf{x}) + \nabla_{\mathbf{x}}(\mathbf{d}^T\mathbf{x}) + \nabla_{\mathbf{x}}(c)$$

$\nabla_{\mathbf{x}}(c) = \mathbf{0} \quad$ (Constant term)
$\nabla_{\mathbf{x}}(\mathbf{d}^T\mathbf{x}) = \mathbf{d} \quad$ (Linear term)

$$\frac{\partial}{\partial x_k}(\mathbf{x}^T\mathbf{Q}\mathbf{x}) = \frac{\partial}{\partial x_k}\left(\sum_{i=1}^{n}\sum_{j=1}^{n} x_i Q_{ij} x_j\right) = \sum_{j=1}^{n} Q_{kj}x_j + \sum_{i=1}^{n} x_i Q_{ik}$$

In matrix:

$$\nabla_{\mathbf{x}}(\mathbf{x}^T\mathbf{Q}\mathbf{x}) = \mathbf{Q}\mathbf{x} + \mathbf{Q}^T\mathbf{x}$$

Since $\mathbf{Q} = \mathbf{Q}^T$:

$$\nabla_{\mathbf{x}}(\mathbf{x}^T\mathbf{Q}\mathbf{x}) = 2\mathbf{Q}\mathbf{x}$$

Combining:

$$\nabla_{\mathbf{x}} J(\mathbf{x}) = 2\mathbf{Q}\mathbf{x} + \mathbf{d}$$

### b) Hessian Matrix $\mathbf{H} = 2\mathbf{Q}$

Defined as the Jacobian of the gradient:

$$\mathbf{H} = \frac{\partial^2 J}{\partial \mathbf{x} \partial \mathbf{x}^T} = \frac{\partial}{\partial \mathbf{x}^T}(2\mathbf{Q}\mathbf{x} + \mathbf{d})$$

Differentiating :

$$\mathbf{H}_{ij} = \frac{\partial}{\partial x_j}\left(\frac{\partial J}{\partial x_i}\right) = \frac{\partial}{\partial x_j}\left(2\sum_{k=1}^{n} Q_{ik}x_k + d_i\right) = 2Q_{ij}$$

$$\mathbf{H} = 2\mathbf{Q}$$

# 4 Question4

## 4.1 Answer

### a) Prediction Formula

$$\hat{y} = \mathbf{x}'_{1\times(p+1)}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

$\mathbf{X}_{n\times(p+1)}$ is the augmented feature matrix
$\mathbf{y}_{n\times 1}$ is the training label vector
$\mathbf{x}'_{1\times(p+1)} = [1, \mathbf{x}'_{1\times p}]$ is the augmented test row vector.

### b) Connection to KNN Regression

Rewrite the prediction as a linear combination of the training labels:

$$\hat{y} = \mathbf{a}^T\mathbf{y} = \sum_{i=1}^{n} a_i y_i$$

where $\mathbf{a}^T = \mathbf{x}'_{1\times(p+1)}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T = [a_1, a_2, \ldots, a_n]$

Exact mathematical structure with distance-weighted KNN regression ($\sum w_i y_i$):

**KNN Regression** : Keeps only the $k$ nearest neighbors and zeroes out the other $n - k$ instances. **Linear Regression** : The special case where $k = n$. It never drops an instance.

# 5 Question5

## 5.1 Answer

To show that $\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$ is a member of the column space of $\mathbf{X}$ (denoted as $\mathcal{C}(\mathbf{X})$), we need to show that $\hat{\mathbf{y}}$ can be expressed as a linear combination of the columns of $\mathbf{X}$.

### a) Matrix Vector Factorization

$$\hat{\mathbf{y}} = \mathbf{X}\left[(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}\right]$$

$$\mathbf{v} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Since $\mathbf{X} \in R^{n\times(p+1)}$ and $\mathbf{y} \in R^n$, the dimension of $\mathbf{v}$ evaluates to a $(p+1)\times 1$ column vector:

$$\mathbf{v} = v_1 v_2 \vdots v_{p+1}$$

$$\hat{\mathbf{y}} = \mathbf{X}\mathbf{v}$$

### b) Linear Combination Form

Denoting the individual columns of $\mathbf{X}$ as $\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_{p+1}$ (where $\mathbf{X} = [\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_{p+1}]$)

$$\hat{\mathbf{y}} = v_1\mathbf{c}_1 + v_2\mathbf{c}_2 + \ldots + v_{p+1}\mathbf{c}_{p+1}$$

### c) Conclusion

Since the column space $\mathcal{C}(\mathbf{X})$ is defined as the set of all possible linear combinations of its columns, it follows that:

$$\hat{\mathbf{y}} \in \mathcal{C}(\mathbf{X})$$

# 6 Question6

## 6.1 Answer

### a) Defining the Optimization

The Residual Sum of Squares (RSS) is defined as:

$$RSS(\beta) = \|\mathbf{y} - \mathbf{X}\beta\|^2 = (\mathbf{y} - \mathbf{X}\beta)^T(\mathbf{y} - \mathbf{X}\beta)$$

Expanding:

$$RSS(\beta) = \mathbf{y}^T\mathbf{y} - 2\beta^T\mathbf{X}^T\mathbf{y} + \beta^T\mathbf{X}^T\mathbf{X}\beta$$

### b) Finding the First-Order Necessary Condition

The gradient with respect to $\beta$ evaluated at $\hat{\beta}$ must be equal to the zero vector:

$$\nabla_\beta RSS(\beta)\Big|_{\beta=\hat{\beta}} = \mathbf{0}$$

$$\nabla_\beta(\mathbf{y}^T\mathbf{y}) = \mathbf{0}$$

$$\nabla_\beta(-2\beta^T\mathbf{X}^T\mathbf{y}) = -2\mathbf{X}^T\mathbf{y}$$

$$\nabla_\beta(\beta^T \mathbf{X}^T \mathbf{X}\beta) = 2\mathbf{X}^T \mathbf{X}\beta$$

Combining these:

$$\nabla_\beta RSS(\beta) = -2\mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \mathbf{X}\beta$$

Setting the gradient to $\mathbf{0}$

$$-2\mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \mathbf{X}\hat{\beta} = \mathbf{0}\mathbf{X}^T \mathbf{X}\hat{\beta} = \mathbf{X}^T \mathbf{y}$$

### c) Orthogonality

$$\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X}\hat{\beta} = \mathbf{0}$$

$$\mathbf{X}^T (\mathbf{y} - \mathbf{X}\hat{\beta}) = \mathbf{0}$$

Substituting $\hat{\mathbf{y}}$ into the parentheses gives:

$$\mathbf{X}^T (\mathbf{y} - \hat{\mathbf{y}}) = \mathbf{0}$$

### d) Conclusion

The matrix equation above directly implies that for every single column $\mathbf{c}_i$:

$$\mathbf{c}_i^T (\mathbf{y} - \hat{\mathbf{y}}) = 0, \quad \forall\, i \in \{1, 2, \dots, p+1\}$$

Since $\mathbf{y} - \hat{\mathbf{y}}$ is orthogonal, the residual vector $\mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to the column space of $\mathbf{X}$.

$$(\mathbf{y} - \hat{\mathbf{y}}) \perp \mathcal{C}(\mathbf{X})$$

# 7 Question7

The instructor suggest using the .arff file, since it preserves higher numerical precision than the .dat file. The divergence in Part (d) is likely due to the lower precision in the .dat data.

## 7.1 a)

Download the Vertebral Column Data
zippath = '/content/drive/MyDrive/USCSummer26/EE559/Data/vertebralcolumndata.zip'
Using Colab

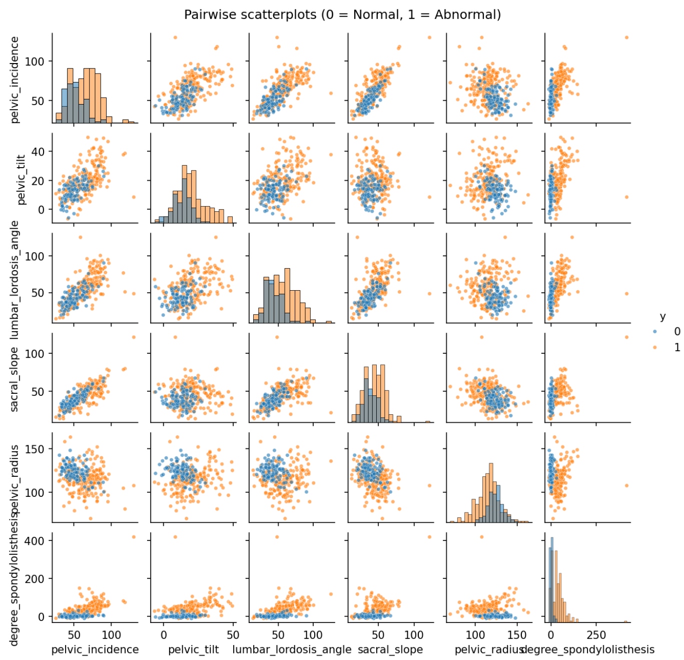


Figure 1: Scatterplots

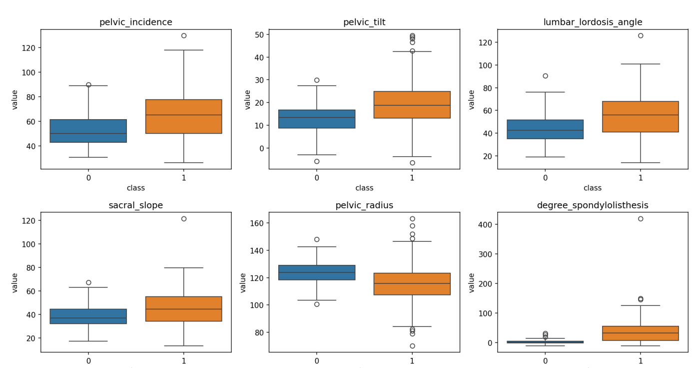


Figure 2: Boxplots

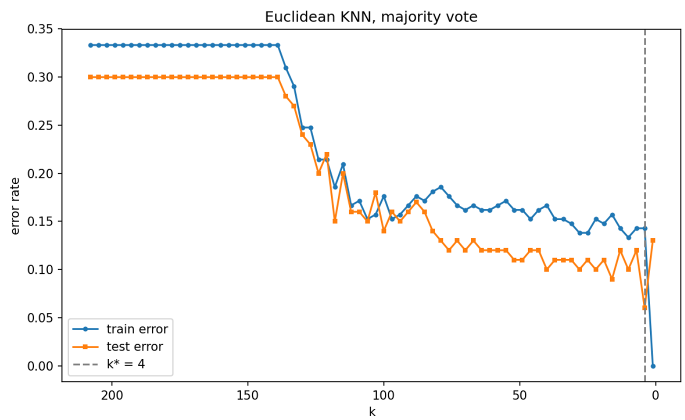


Figure 3: Training and test error rates in terms of $k$ for Euclidean distance metric (majority voting). The dashed line marks the optimal $k^* = 4$.

### 7.2 b)

**i Scatterplots: As Figure 1 shows.**

**ii Boxplots: As Figure 2 shows**

**iii Data Splitting:**

**Training Set** ($N = 210$)**:** 70 rows of Class 0, 140 rows of Class 1.
**Test Set** ($N_{test} = 100$)**:** 30 rows of Class 0, 70 rows of Class 1.

### 7.3 c)

**(c) i.** Use `sklearn.neighbors.KNeighborsClassifier` with Euclidean distance

```
from sklearn.neighbors import KNeighborsClassifier

def knn_errors(Xtr, ytr, Xte, yte, k):
    clf = KNeighborsClassifier(n_neighbors=k,
              metric='euclidean', weights='uniform')
    clf.fit(Xtr, ytr)
    tr_err = np.mean(clf.predict(Xtr) != ytr)
    te_err = np.mean(clf.predict(Xte) != yte)
    return tr_err, te_err
```

**(C)ii.**

The most suitable neighbor size is found to be $k^* = 4$, yielding a minimum test error rate of 0.0600 (6.0 %).

**Confusion Matrix** The confusion matrix at $k^* = 4$

| | | Predicted | |
|---|---|---|---|
| | | Abnormal | Normal |
| 2***Actual** | Abnormal | 69 (TP) | 1 (FN) |
| | Normal | 5 (FP) | 25 (TN) |

**Performance Summary**

**True Positive Rate (TPR):** 0.986 (98.6 %)
**True Negative Rate (TNR):** 0.833 (83.3 %)
**Precision:** 0.932 (93.2 %)
$F_1$**-Score:** 0.958

## 7.4 d)

Table 1 compares the optimal $k$ and minimum test error rates for each distance metric using majority polling ($k \in \{1, 6, 11, \ldots, 196\}$).

## 7.5 e)

The results for $k \in \{1, 6, 11, \ldots, 196\}$ are presented in Table 2.

## 7.6 f)

The lowest training error rate achieved in this homework is **0.0000** (0.0 %), which is uniquely achieved when $k = 1$.

*When $k = 1$, each training sample's nearest neighbor is itself. This creates an identity mapping.*

Table 1: Comparison of best test errors.

| **Distance Metric** | **Optimal $k$** | **Best Test Error Rate** |
|---|---|---|
| **Euclidean** | 6 | **0.0800** |
| **Chebyshev** | 16 | **0.0800** |
| **Manhattan** ($p = 1$) | 1 | 0.1100 |
| **Minkowski** ($\log_{10}(p) = 0.7$) | 1 | 0.1100 |
| **Mahalanobis** | 1 | 0.1500 |

Table 2: Comparison of best test errors

| **Distance Metric** | **Optimal $k$** | **Best Test Error Rate** |
|---|---|---|
| **Euclidean** | 6 | **0.1000** |
| **Manhattan** | 26 | **0.1000** |
| **Chebyshev** | 16 | 0.1100 |